



Lecture 2

User-centric approaches

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Goals of this lecture

1

Distinguish between platform- and user-centric approaches to digital trace data collection.

2

Understand how commonly used user-centric approaches work and how they can be used for research.

3

Evaluate the advantages, challenges and ethical considerations of user-centric approaches.

4

Apply by learning from example studies that show how research questions can be answered with user-centric approaches.

Literature

- **WAS chapter 1:** Wearables, apps and sensors as novel tools for data collection
- **Boeschoten et al. (2022):** A framework for privacy preserving digital trace data collection through data donation
- **Ram et al. (2019): Screenomics:** A new approach for observing and studying individuals' digital lives
- **Hendrickx (2025):** News #foryou on TikTok: A digital methods-based study

Outline

1

Recap: User- versus platform-centric approaches. *Why a user-centric approach?*

2

Discuss: three common user-centric approaches with examples

3

Evaluate: the methods in light of advantages, challenges and ethical considerations

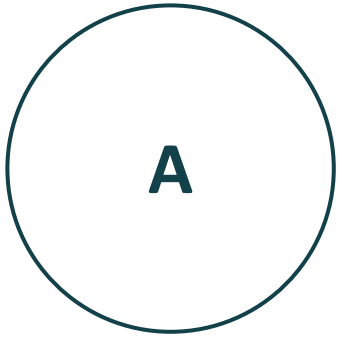
4

Consider: The new study designs in the context of found and designed data

01

Recap

User-centric versus platform-centric approaches



Definitions

Digital Traces

- Also referred to as “Digital Footprint” or “Digital Shadow”
- Definition: *“One's unique set of digital activities, actions, and communications that leave a data trace on the internet or on a computer or other digital device and can identify the particular user or device.”*

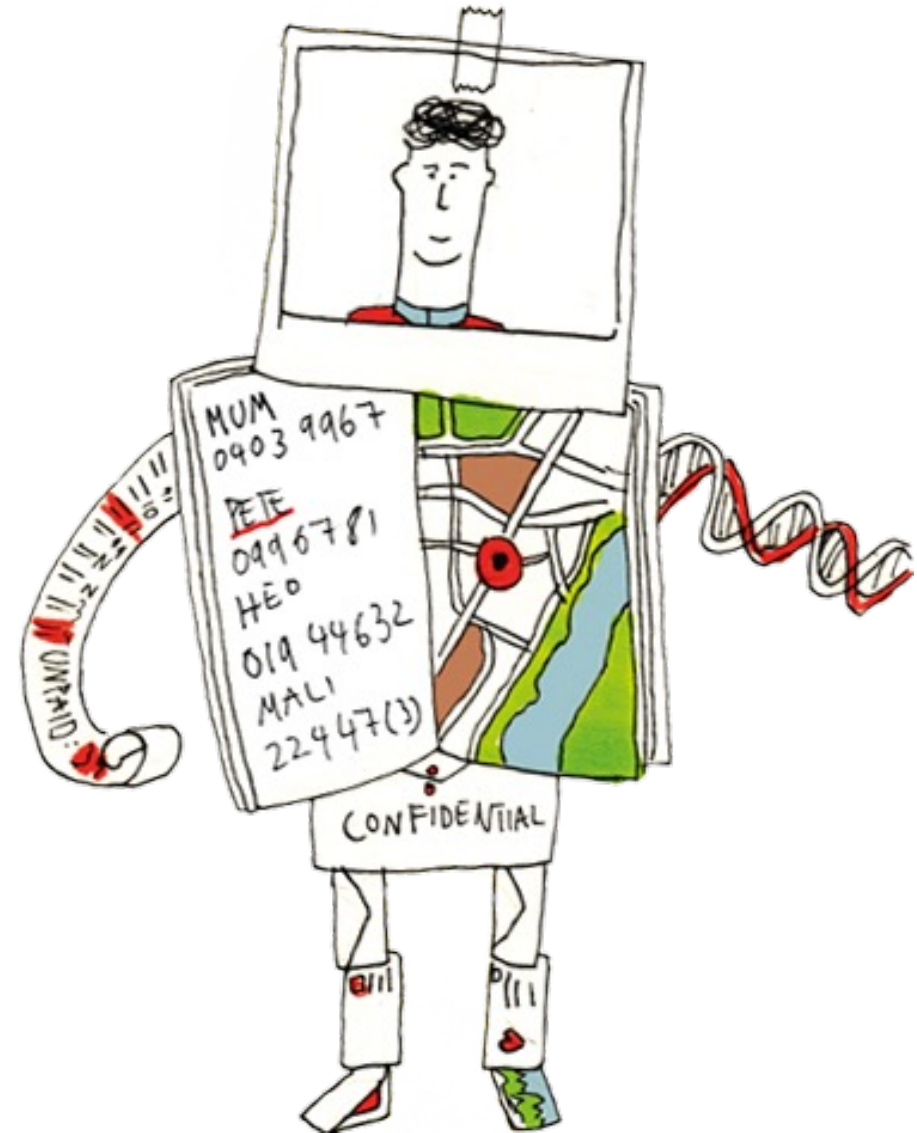
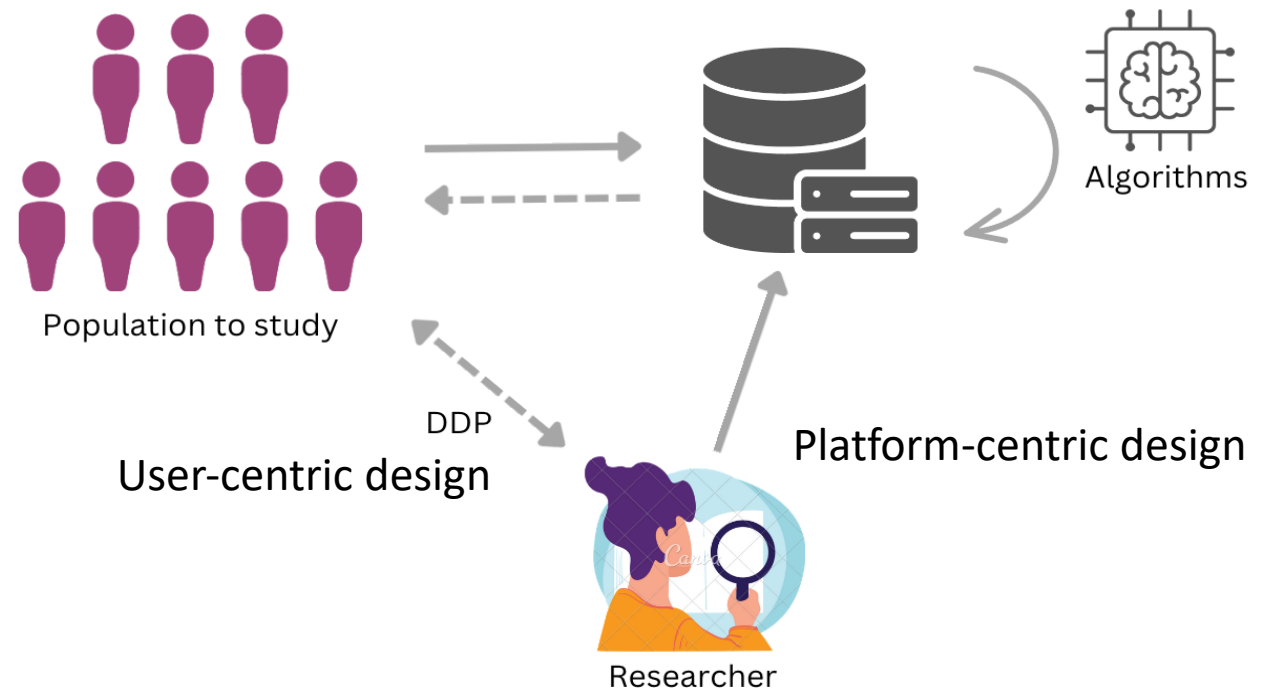


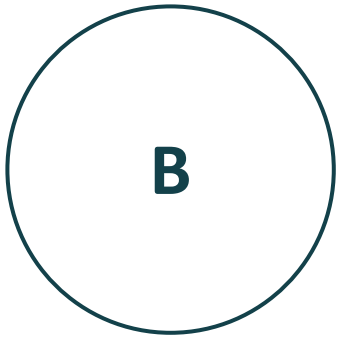
Image: <https://myshadow.org/tracking-data-traces>

Definition: <https://www.dictionary.com/browse/digital-footprint>

User-centric and platform-centric

1. You are interested in making inferences about a population.
2. You are interested in concepts that can be measured through digital traces.
3. Platform centric → get the traces from the platform directly
4. User centric → collaborate with a sample of that population directly to obtain access to their digital traces





Reasons to take a user-centric approach

Reasons to take a user-centric approach

New data types

- Allows for person-specific analysis (Valkenburg et al., 2021)
- Allows to capture private / non-public data (Ohme et al., 2023)

Relation to platforms

- Allows researchers to overcome restrictions on APIs and scraping (Bruns, 2019; Breuer et al., 2023)
- Avoids platform dependency (Ohme et al., 2023)

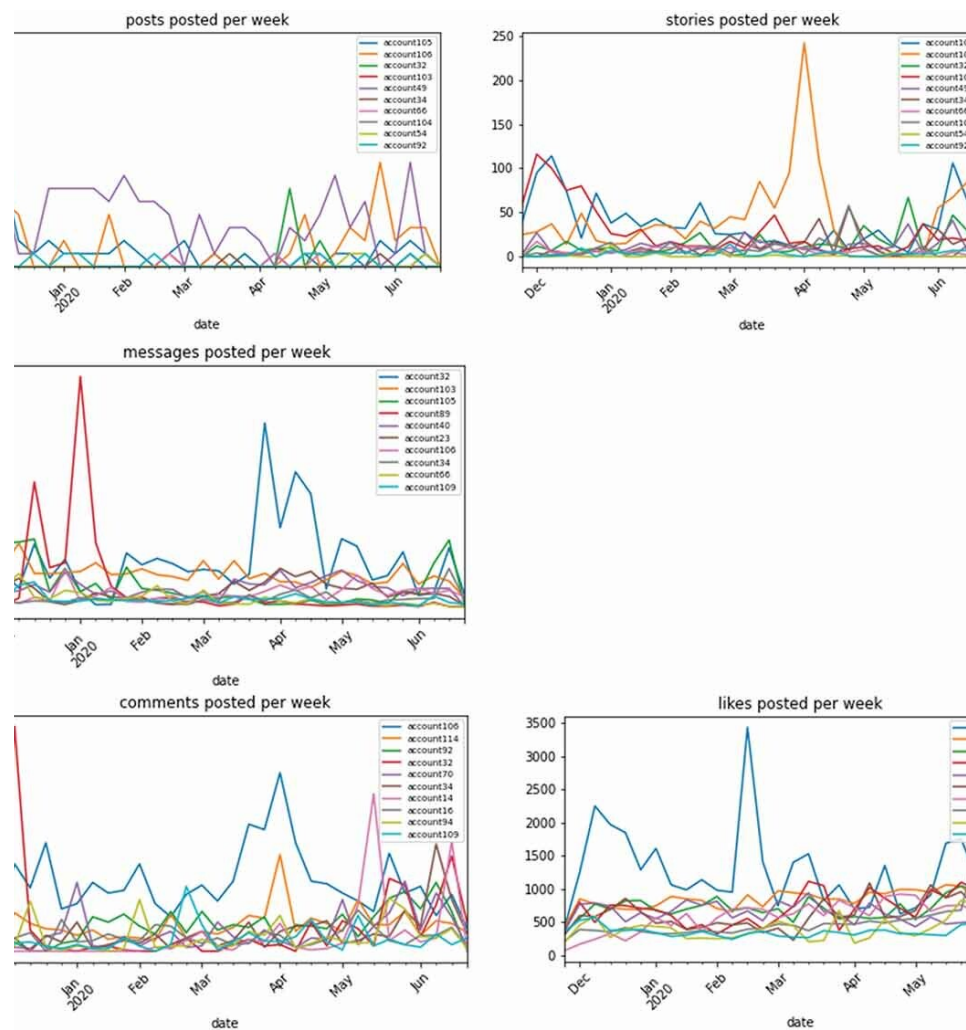
Role of participants

- Allows to obtain active informed consent (de León et al., 2026)
- Allows to define your population and recruit your sample yourself (Breuer et al., 2023; Ohme et al., 2023)

Data quality

- Allows to combine digital traces with other data sources on person level (Ohme et al., 2023)
- Reduces the reliance on self-reports (de León et al., 2026)

Illustrated by Project AweSome



Measurement Burst Design (Nesselrode, 1991)

N = 388 (age 13-15)

T > 250

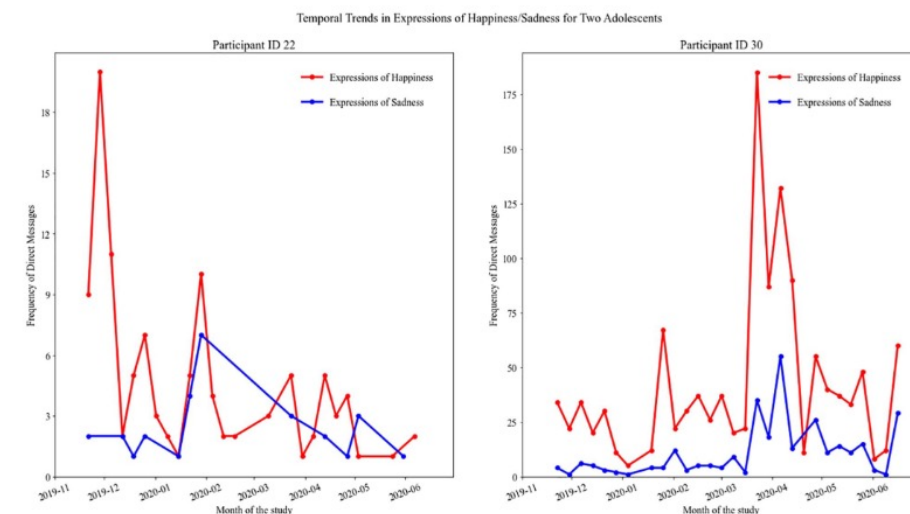


Figure 4. Temporal trends in the frequency of direct messages containing expressions of happiness (red line) and sadness (blue line) for two different participants.

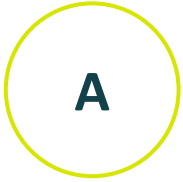
Note. The y-axis differs per plot.



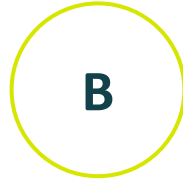
02

Three common user-centric approaches

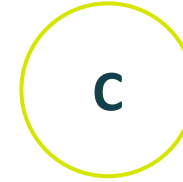
Three common user-centric approaches



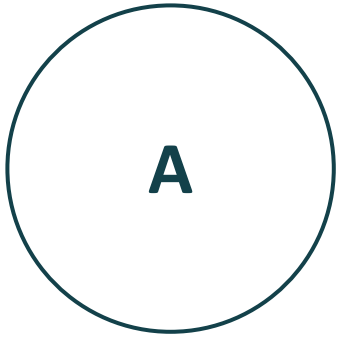
Sensor approaches



Capture and scraping approaches



Data donation approaches



Sensor approaches

1

Definitions

2

In practice

3

Example

4

Exercise

Sensor approaches - definitions

DEFINITION

Sensor

A piece of hardware that converts a physical phenomenon into an electric signal.

For example: the accelerometer sensor measures the rate of change of velocity of an object

DEFINITION

Wearable

A lightweight device including one or multiple sensors that people can wear close to or on their skin.

For example: a fitness bracelet or smartwatch.

DEFINITION

App

A software program that runs on a smartphone or other smart device. Apps allow users to interact with a device.

For researchers, apps provide the opportunity to collect, aggregate and process data collected on the device.

1

Definitions

2

In practice

3

Example

4

Exercise

Sensor approaches – more definitions

DEFINITION

Passive data collection

Data are collected, for example by sensors, as a byproduct of an everyday activity.

For example: an accelerometer collects data while the participant walks.

DEFINITION

Active data collection

A participant deliberately creates data as a reaction to a stimulus designed by the researcher.

For example: the participant responds to a survey question to take a picture and share it with the researcher.

1

Definitions

2

In practice

3

Example

4

Exercise

How does it work in practice? *Choose between these alternatives*

A

Direct sensor use

You ask participants to use the sensor directly

B

Custom software

You develop a piece of software (e.g. app or plug-in) that captures the traces you are interested in

C

Use existing software

You make use of an existing app or plug-in

1

Definitions

2

In practice

3

Example

4

Exercise

How does it work in practice? *Procedure*

1

Invite participants to your study

2

**Instruct them to use [A, B or C from
the previous slide]**

3

**From that moment, data collection
happens prospective**

*This means: Digital traces are collected as they are
being produced.*

1

Definitions

2

In practice

3

Example

4

Exercise

What sensors can you use?

NFC

Thermometer

Wi-Fi

Cellular network

Bluetooth

Air humidity sensor

Light sensor

Camera

Proximity sensor

Microphone

Fingerprint sensor

Compass

Barometer

Accelerometer

Pedometer

Gyroscope

1

Definitions

2

In practice

3

Example

4

Exercise

What can you measure with these sensors?

NFC

Thermometer

Wi-Fi

Cellular network

Bluetooth

Air humidity sensor

Light sensor

Camera

Proximity

Proximity sensor

Microphone

Fingerprint sensor

Compass

Barometer

Accelerometer

Pedometer

Gyroscope

Physical activity

Location

Ambience

1

Definitions

2

In practice

3

Example

4

Exercise

Sensor categories

Location

- GPS
- Cellular network
- Wi-Fi
- Beacons

Physical activity

- Accelerometer
- Gyroscope
- Barometer
- Pedometer

Ambience & proximity

- Camera
- Microphone
- Light sensor
- Bluetooth
- NFC

1

Definitions

2

In practice

3

Example

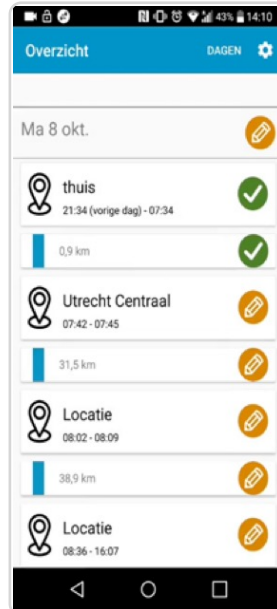
4

Exercise

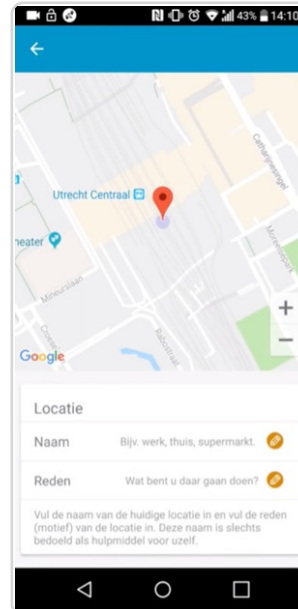
EXAMPLE

Measuring everyday mobility using an app

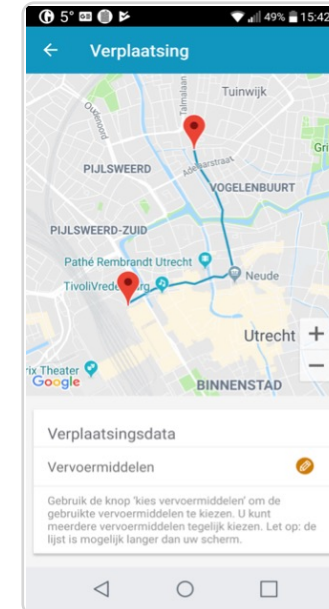
N = 1,902 · Dutch general population, field test Nov–Dec 2018: Statistics Netherlands' TABI app (Android & iOS), 7 days, GPS + Wi-Fi



Daily overview



Questions about stops



Questions about trips

Definitions

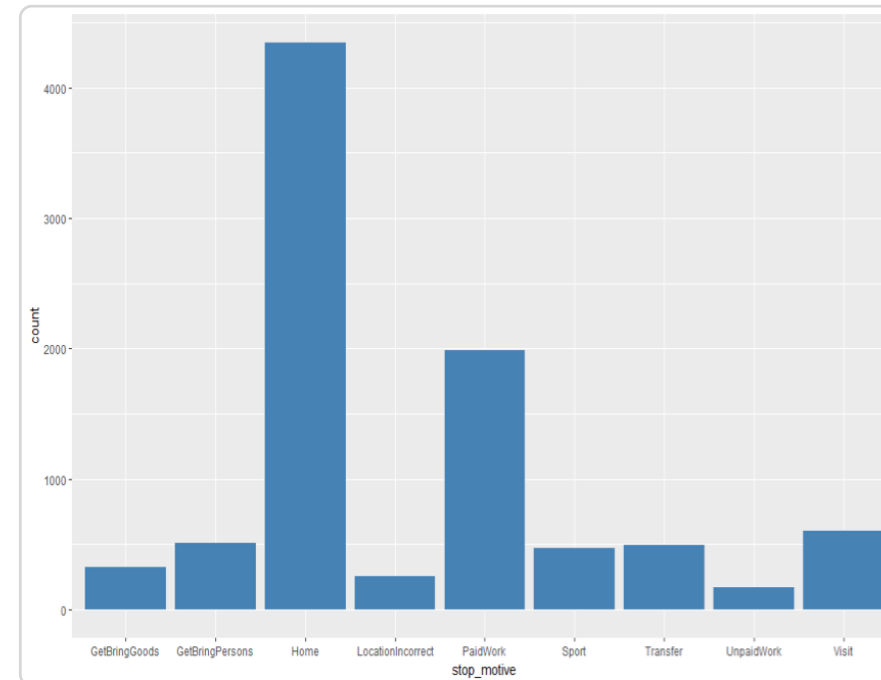
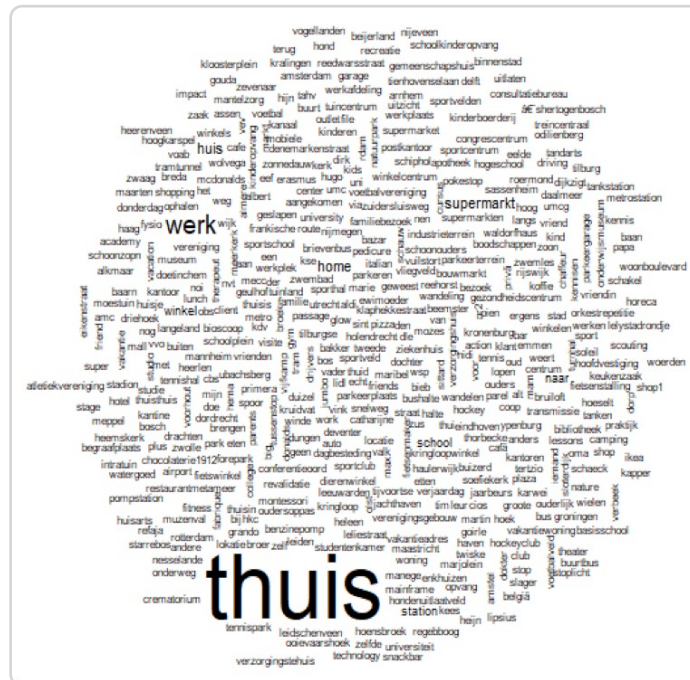
Motives and means of transportation

22,000 stops logged · about **50%** of respondents added compliant detail on their trip

In practice

Example

Exercise



Stop labels (word cloud, in Dutch)

Stop motives (count)

1

Definitions

2

In practice

3

Example

4

Exercise

EXERCISE

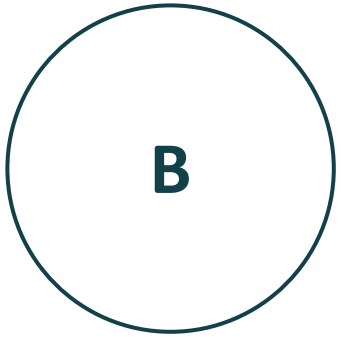
What sensors do you have on you? And can it be used for research?

1

Together with your neighbor, list every sensor you have on you (e.g. on your phone, watch, earbuds, laptop)

2

Pick one sensor. What could you measure with this? What could go wrong when doing this for research? (e.g. ethically or practically)



Capture and scraping approaches



1

Definitions

2

In practice

3

Example

4

Exercise

Capture and scraping approaches — definitions

DEFINITION

Screen capture

A software application installed on the participant's device that records what's on their screen without relying on the phone's own sensors. This can happen continuously or as periodic screenshots.

DEFINITION

Scraping

A software application that automatically collects and extracts specific content (text, images, page structure) from digital sources such as an app or a website. It does not create a full visual copy of the screen.

1

Definitions

2

In practice

3

Example

4

Exercise

How does it work in practice?

1

Invite

You invite participants to your study.

2

Install

They install an app or plug-in that captures their screen or scrapes specific content.

3

Collect

Data collection is prospective: screenshots or scraped content are logged automatically as they use their device.

1

Definitions

2

In practice

3

Example

4

Exercise

EXAMPLE

The human screenome project

Screenomics software records, encrypts and transmits screenshots automatically and unobtrusively

every 5 seconds, whenever a device is turned on.

When deployed on multiple devices at once, the screenshots from each one are synced in time.

Palo Alto Transit Center (Caltrain Platform South, Drop-off)	Palo Alto Transit Center (Caltrain Platform South, Pick-up)	San Jose (Quarry)
12:45 AM	1:00 AM	1:15 AM
1:00 AM	1:15 AM	1:30 AM
1:15 AM	1:30 AM	1:45 AM
1:30 AM	1:45 AM	2:00 AM
1:45 AM	2:00 AM	2:15 AM
2:00 AM	2:15 AM	2:30 AM
2:15 AM	2:30 AM	2:45 AM
2:30 AM	2:45 AM	3:00 AM
2:45 AM	3:00 AM	3:15 AM
3:00 AM	3:15 AM	3:30 AM
3:15 AM	3:30 AM	3:45 AM
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11:45 PM	12:00 AM	12:15 AM
12:00 AM	12:15 AM	12:30 AM
12:15 AM	12:30 AM	12:45 AM

Screenshots collected for one participant

1

Definitions

2

In practice

3

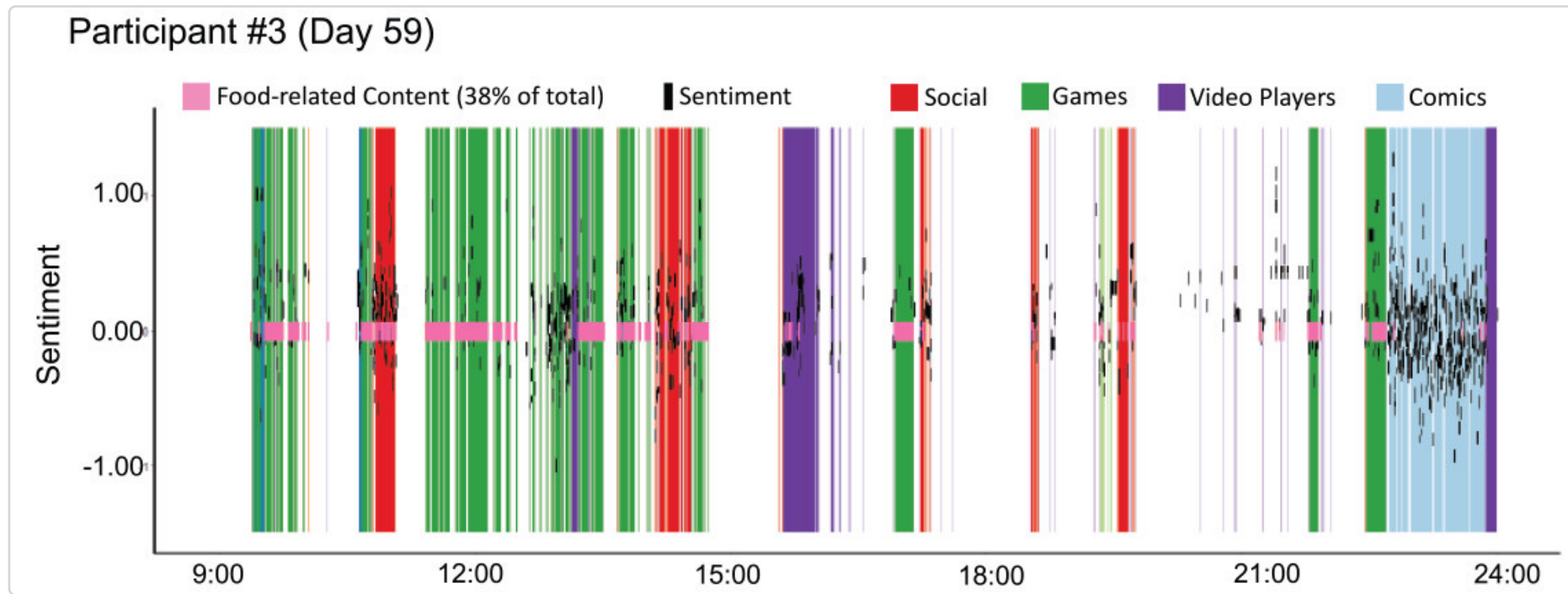
Example

4

Exercise

EXAMPLE

Example data from the Screenomics app



Participant #3, Day 59: App category and sentiment reconstructed from screenshots taken every 5 seconds

1

Definitions

2

In practice

3

Example

4

Exercise

EXERCISE

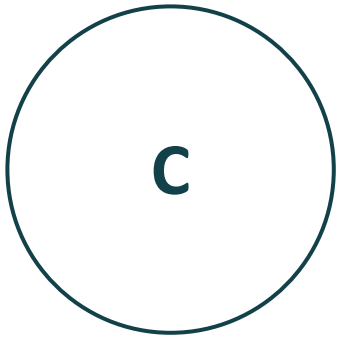
Discuss with your neighbor: If a screenshot is made every 5 seconds ...

1 What would this reveal about you that GPS or step count never could?

2 Think about what your Screenome from yesterday would have looked like. What is one thing that would be surprising about this?

15

MINUTE COFFEE BREAK



Data donation approaches

1

Definitions

2

In practice

3

Example

4

Exercise

GDPR – definitions

DEFINITION

Personal data

Information relating to an identified or identifiable natural person

DEFINITION

Data subject

The natural person that the personal data refer to.

DEFINITION

Data controller

The natural person or organization responsible for processing personal data. The controller decides which data will be processed, how and why.



1

Definitions

2

In practice

3

Example

4

Exercise

Data donation — definitions

DEFINITION

Data Download Package (DDP)

Under GDPR's right of access and data portability, platforms must provide users their personal data in a machine-readable format, typically a .zip file.

DEFINITION

Local processing

Extracting only the data that is relevant to a research question directly on the participant's own device, before anything is shared with the researcher.

1

Definitions

2

In practice

3

Example

4

Exercise

How does it work in practice?

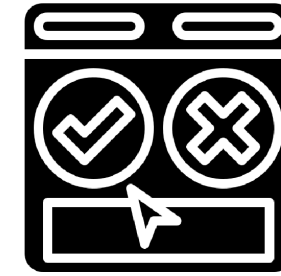
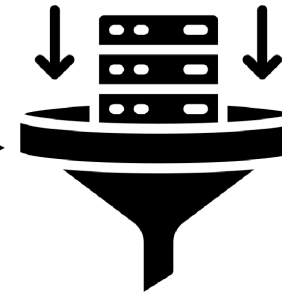
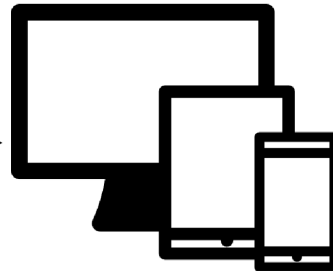
Informed consent

Data Download Packages (DDPs)

Local processing

Inspect and donate

Data analysis



Participant

Digital trace data

Participant's device

Variables of interest

Researcher

1

Request

The participant requests their DDP directly from the platform.

2

Review & select

They open the package locally and choose what to donate.

3

Donate

The selected data is submitted to the researcher; the rest stays on the participant's device.

Two ways to do data donation: share the complete DDP, or process it locally and share only what's relevant. Local processing is now the more common approach.

1

Definitions

2

In practice

3

Example

4

Exercise

EXAMPLE 1

How common is binge-watching?

WHAT'S IN THE NETFLIX DDP

ViewingActivity.csv

Profile	Start time	Dur.	Title
Alex	2024-03-02 21:14	42:10	Stranger Things S3:E4
Alex	2024-03-02 21:58	38:22	Stranger Things S3:E5
Alex	2024-03-02 22:37	40:05	Stranger Things S3:E6
Jamie	2024-03-03 08:12	12:44	Bluey S2:E1
Jamie	2024-03-03 19:41	51:30	The Crown S4:E2

AFTER PROCESSING

MIN. ITEMS IN A SESSION	% BINGE
Two items	32.2%
Three items	14.8%
Four items	8%
Five items	4.7%
Six items	2.8%

1

Definitions

2

In practice

3

Example

4

Exercise

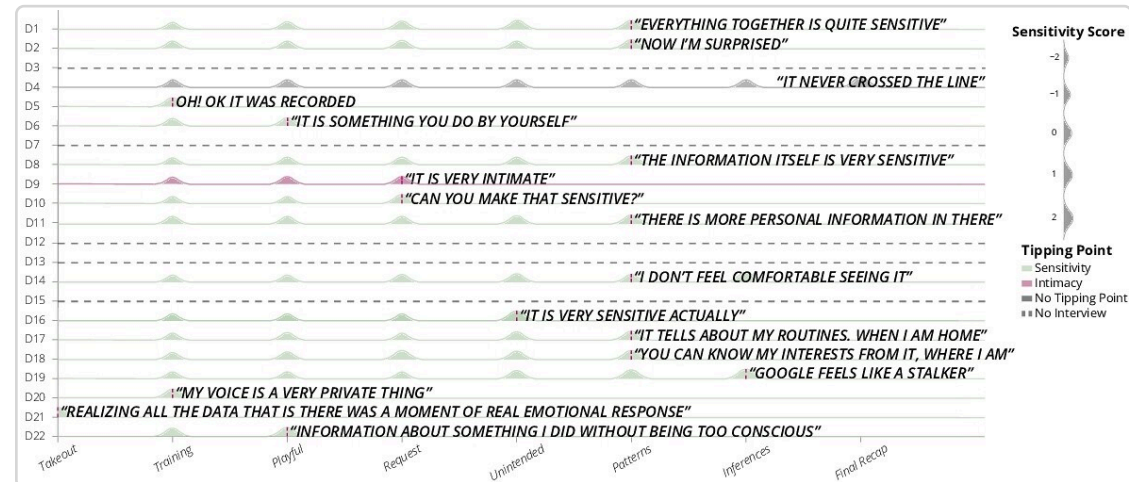
EXAMPLE 2

Sensitivity and intimacy with Google Assistant users

WHAT'S IN THE DDP

MyActivity.json	
Time	Query (verbatim)
14:02	"Remind me to call mom at 5"
14:15	"What's the weather tomorrow"
19:41	"Play calming sleep sounds"
21:12	"Am I stressed, how can I tell"
21:13	"How to stop feeling anxious"

AFTER PROCESSING



1

Definitions

2

In practice

3

Example

4

Exercise

EXERCISE

A researcher wants to use TikTok data donation for the RQ: “Does short-form video use before bed predict how well someone sleeps?”

1

What is in the raw TikTok DDP? (you cannot look it up directly, but you can guess...)

2

How could you process this into measurements that answer the research question?

3

What can be done locally on the participant’s device?

03

Advantages, challenges, ethical considerations

Advantages

All approaches

- Naturalistic context: behavior is observed as it happens, not staged for the study
- Large scale: across many participants or data points
- Detailed, granular data
- Captures concepts that were previously hard to measure
- Can be supplemented with other data sources, e.g. surveys

Sensor-based

- Relatively low participant effort: Sensors run passively in the background

Capture / scraping

- Relatively low participant effort: Once installed, the tool runs unobtrusively

Data donation

- Data is collected retrospectively.
- Not influenced by participants “feeling seen” while it's collected
- Relatively cheap to set up: No custom app or sensor deployment needed (only potentially set-up local processing)

Challenges

All approaches

- Not everyone can, or wants to, participate
- Substantial data processing needed before data is usable
- Non-participation due to privacy concerns
- Participants need to have digital skills

Sensor-based

- App or sensor development is expensive
- Expensive to maintain over time

Capture / scraping

- Some platforms actively block scraping
- Requires active maintenance as platforms change

Data donation

- Substantive effort required from participants: Requesting and reviewing their own data
- Every study needs a custom design; no one-size-fits-all pipeline

Ethical considerations

All approaches

- Data passes through multiple stages: Carefully consider who has access at each stage

Sensor-based

- Participants may forget they are being observed over time

Capture / scraping

- Participants may forget they're being observed (as in screenomics)
- Information of contacts may be revealed incidentally

Data donation

- Participants can inspect data before sharing → but can they really oversee such large volumes of data?
- Contacts' information may be revealed too. But local processing can help prevent this

EXERCISE

Pick a method for this research question:

“How does exposure to news on social media affect people's mood throughout the day?”

- 1 **Choose a method:** Sensor based, capture/scraping, or data donation?
- 2 **Explain why:** Name one advantage that makes it the right fit, and one challenge or ethical risk you'd need to manage.

04

Found and designed data

Found data, designed data

From a researcher's perspective, big data sources look **"found"**: already out there, waiting to be used. But someone **"designed"** it first.

DEFINITION

Found data

Data that already exists “out there”, which researchers didn't set out to create. Examples are platform logs, transaction records, social media posts. It was already there, before it was collected for a (research) purpose.

DEFINITION

Designed data

The same data, seen from another angle: every field, format, and default was chosen by someone (e.g. a platform, a research team) for a reason of their own. This means that no aspect of the data is neutral or automatic.

Two questions to ask of any data source

1

Active or passive?

Was this data actively created by the person (e.g. typing a message, writing a post) or passively logged as a by-product of just using a device or platform?

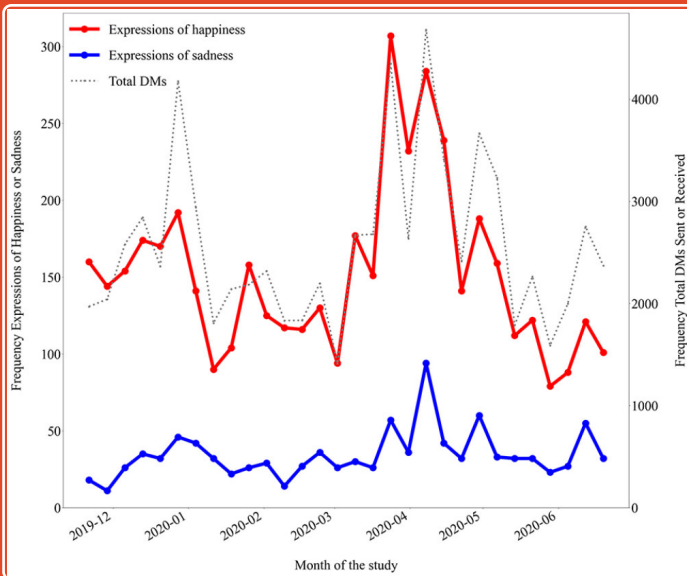
2

Who designed the data, and why?

Every field that got captured, every option offered, every default that was set has been decided on by someone for their own reasons. What were their incentives to shape the data like this?

EXERCISE

Look again at Project AweSome



Happiness and sadness in adolescents' Instagram DMs

See Verbeij et al. (2024)

1

Was this data collection active or passive?

Different aspects: messages, sentiment scores ...

2

Who designed the data, and why?

E.g. think of the DM feature of Instagram.



Questions?

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